



Date of Birth: **13 March 1999**|Gender: **Male**| Marital Status: **Single**

EDUCATION

- Sep.2023- Jul.2025**
Master of Computer Engineering- Software
Islamic Azad University, Tabriz Branch, Iran
GPA: 19.25 (out of 20)
Thesis Title: Toward Sustainable Healthcare Process Mining: A Polyglot Hybrid Database Architecture for Efficient Analytics and CO₂ Reduction
Supervisor: Dr. Mohammadbagher Karimi
Course Focus: Advanced Software Engineering, Evaluating Performance of Computer Systems, Methodology of Software Production, Distributed Systems, Applied Machine Learning, Requirements Engineering
- Sep.2018- Jul.2022**
Bachelor of Computer Engineering
Islamic Azad University, Tabriz Branch, Iran
Last two years' GPA: 16.38 (out of 20)
Final Project: Parent Connect - designed and developed a Django-based social platform enabling parents to collaborate, share insights, and exchange feedback on their children's educational and developmental experiences
Supervisor: Dr. Mohammadbagher Karimi
Course Focus: Information Technology Project Management, Microprocessor & Assembly Language, Fundamentals of Safe Computing, Engineering Economics, Principles of Computer Design

RESEARCH INTERESTS

- Process Mining/Healthcare Process Mining
- AI/ML-Driven Software Defect Prediction
- Sustainable & Polyglot Database Architecture Design
- Agentic AI & Intelligent Systems in Software Engineering
- Data-Intensive Software System Design

PUBLICATIONS

Journal Articles

- Jul. 2026**
Authors: Aydin Fakhri Kaleybar, Mohammadbagher Karimi, Niloofar Naghshi, Bahman Arasteh
Title: Systematic Review: A Literature Review of Process Mining in Healthcare, *International Journal of Data Science and Analytics*, Springer, [Link](#)

Under Review Articles

- Authors:** Niloofar Naghshi, Mohammadbagher Karimi, Aydin Fakhri Kaleybar, Bahman Arasteh, Huseyin Kusetogullar
Title: Using Advanced Learning Techniques to Predict Software Defects: A Systematic Literature Review, *International Journal of Computational Intelligence Systems*, Springer Nature

In preparation Manuscripts

- Authors:** Aydin Fakhri Kaleybar, Mohammadbagher Karimi, Niloofar Naghshi, Bahman Arasteh, Huseyin Kusetogullar, Asgharali Bouyer
Title: Agentic AI and Methodological Transformation of Software Engineering: A Systematic Review of Intelligent Agents in the Development Lifecycle
- Authors:** Aydin Fakhri Kaleybar, Mohammadbagher Karimi, Bahman Arasteh, Niloofar Naghshi, Huseyin Kusetogullar
Title: Toward Sustainable Healthcare Process Mining: A Polyglot Hybrid Database Architecture for Efficient Analytics and CO₂ Reduction
- Authors:** Niloofar Naghshi, Bahman Arasteh, Mohammadbagher Karimi, Aydin Fakhri Kaleybar, Huseyin Kusetogulla, Asgarali Bouyer
Title: Parrot Optimizer: A Systematic Review of Algorithmic Evolution, Hybridization Strategies, Application Domains, and Future Directions

ACADEMIC EXPERIENCES

- **Jul. 2025- Present**
Independent Research Collaboration with supervisor, Islamic Azad University, Tabriz Branch, Iran
- **Nov. 2023- Jul. 2025**
Research Assistant, Islamic Azad University, Tabriz Branch, Iran

CERTIFICATES

- **Dec. 2023**
Advancing the Digital Economy for Sustainable Growth in Asia, granted by ADB Institute
- **Dec. 2023**
Exam Prep: AWS Certified Cloud Practitioner Foundations, granted by Coursera

LANGUAGE PROFICIENCY

- **Mother Tongue** Persian (Farsi) and Azerbaijani Turkish
- **Other languages** English (Advanced C1), Istanbul Turkish (Upper Intermediate B2)

SKILLS

- **Job-related Software Skills**
Professional: Python, FastAPI, MySQL, Git, Django, Redis, PostgreSQL
Intermediate: MongoDB, Pandas, Scikit-learn, Matplotlib, Docker, ProM
Familiar: Disco, Alpha Miner, Heuristic Miner, Inductive Miner, TimescaleDB, ChromaDB, Neo4j
- **General Software Skills**
Professional: Microsoft Word, Microsoft PowerPoint, Windows OS
Intermediate: Linux OS, Google Workspace (Docs/Sheets), Microsoft Excel
- **Soft Skills**
Strong Communication Skills, Analytical Thinking, Problem Solving, Academic/Scientific Writing, Effective Communication, Time Management, Attention to Detail, Critical Thinking, Independent/Self-directed Learning, Research Project Management, Adaptability, Teamwork Skills

REFERENCES

- **Referee 1** **Dr. Mohammadbagher Karimi**
- **Affiliation** Assistant Professor at Islamic Azad University, Tabriz Branch (IAUT), Iran
- **Email** mba.karimi@iau.ac.ir
- **Referee 2** **Dr. Bahman Arasteh**
- **Affiliation** Associate Professor at Istinye University, Istanbul, Turkey
- **Email** bahman.arasteh@istinye.edu.tr
- **Referee 3** **Dr. Asgarali Bouyer**
- **Affiliation** Associate Professor at Azarbaijan Shahid Madani University, Iran
- **Email** a.bouyer@azaruniv.ac.ir

HONORS

- Being among the top three students in the master of Computer Engineering with a GPA of 19.25/20
- Received a score of 20/20 for the final thesis defense at the master's level
- Contribution to five research projects in prestigious Journal with one article as the first author and others under prep and revision